

Auteur: Stan Van Campenhout

Studierichting: Informatica

Oefeningenreeks 7B nr. 6.1

$$f : \mathbb{R}^3 \rightarrow \mathbb{R} : (x, y, z) \rightarrow xyz$$

$$g : \mathbb{R} \rightarrow \mathbb{R}^2 : x \rightarrow (2x, x^3)$$

$$h : \mathbb{R}^2 \rightarrow \mathbb{R}^3 : (x, y) \rightarrow (x^2, y^2, x^2y)$$

$$Dh = \begin{pmatrix} 2x & 0 \\ 0 & 2y \\ 2xy & x^2 \end{pmatrix}$$

$$Dg = \begin{pmatrix} 2 \\ 3x^2 \end{pmatrix}$$

$$Df = \begin{pmatrix} yz & xz & xy \end{pmatrix}$$

$$D(h \circ g \circ f) = Dh(g(f(x))) \cdot Dg(f(x)) \cdot Df(x)$$

$$Df(x) = \begin{pmatrix} yz & xz & xy \end{pmatrix}$$

$$Dg(f(x)) = \begin{pmatrix} 2 \\ 3(xyz)^2 \end{pmatrix}$$

$$Dh(g(f(x))) = \begin{pmatrix} 4xyz & 0 \\ 0 & 2(xyz)^3 \\ 4(xyz)^4 & 4(xyz)^2 \end{pmatrix}$$

$$\Rightarrow D(h \circ g \circ f)(x) = \begin{pmatrix} 4xyz & 0 \\ 0 & 2(xyz)^3 \\ 4(xyz)^4 & 4(xyz)^2 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 3(xyz)^2 \end{pmatrix} \cdot \begin{pmatrix} yz & xz & xy \end{pmatrix}$$

$$= \begin{pmatrix} 8xyz \\ 6(xyz)^5 \\ 20(xyz)^4 \end{pmatrix} \cdot \begin{pmatrix} yz & xz & xy \end{pmatrix}$$

$$= \begin{pmatrix} 8xy^2z^2 & 8x^2yz^2 & 8x^2y^2z \\ 6x^5y^6z^6 & 6x^6y^5z^6 & 6x^6y^6z^5 \\ 20x^4y^5z^5 & 20x^5y^4z^5 & 20x^5y^5z^4 \end{pmatrix}$$

$$D(g \circ f \circ h)(x) = Dg(f(g(x))) \cdot Df(h(x)) \cdot Dh(x)$$

$$Dh(x) = \begin{pmatrix} 2x & 0 \\ 0 & 2y \\ 2xy & x^2 \end{pmatrix}$$

$$Df(h(x)) = \begin{pmatrix} x^2y^3 & x^4y & x^2y^2 \end{pmatrix}$$

$$Dg(f(g(x))) = \begin{pmatrix} 2 \\ 3(x^4y^3)^2 \end{pmatrix}$$

$$\Rightarrow D(g \circ f \circ h) = \begin{pmatrix} 2 \\ 3(x^4y^3)^2 \end{pmatrix} \cdot \begin{pmatrix} x^2y^3 & x^4y & x^2y^2 \end{pmatrix} \cdot \begin{pmatrix} 2x & 0 \\ 0 & 2y \\ 2xy & x^2 \end{pmatrix}$$

$$= \begin{pmatrix} 2x^2y^3 & 2x^4y & 2x^2y^2 \\ 3x^{10}y^9 & 3x^12y^7 & 3x^{10}y^8 \end{pmatrix} \cdot \begin{pmatrix} 2x & 0 \\ 0 & 2y \\ 2xy & x^2 \end{pmatrix}$$

$$= \begin{pmatrix} 8x^3y^3 & 6x^4y^2 \\ 12x^{11}y^9 & 9x^{12}y^8 \end{pmatrix}$$

$$D(f \circ h \circ g)(x) = Df(h(g(x))) \cdot Dh(g(x)) \cdot Dg(x)$$

$$Dg(x) = \begin{pmatrix} 2 \\ 3x^2 \end{pmatrix}$$

$$Dh(g(x)) = \begin{pmatrix} 4x & 0 \\ 0 & 2x^3 \\ 4x^4 & 4x^2 \end{pmatrix}$$

$$Df(h(g(x))) = \begin{pmatrix} 4x^{11} & 16x^7 & 4x^8 \end{pmatrix}$$

$$\Rightarrow D(f \circ h \circ g)(x) = \begin{pmatrix} 4x^{11} & 16x^7 & 4x^8 \end{pmatrix} \cdot \begin{pmatrix} 4x & 0 \\ 0 & 2x^3 \\ 4x^4 & 4x^2 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 3x^2 \end{pmatrix}$$

$$= \begin{pmatrix} 4x^{11} & 16x^7 & 4x^8 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 3x^2 \end{pmatrix} = 64x^{12} + 144x^{12} = 208x^{12}$$

References